

**МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ
РОССИЙСКОЙ ФЕДЕРАЦИИ
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ
ВЫСШЕГО ОБРАЗОВАНИЯ
«БЕЛГОРОДСКИЙ ГОСУДАРСТВЕННЫЙ
ТЕХНОЛОГИЧЕСКИЙ УНИВЕРСИТЕТ им. В. Г. ШУХОВА»
(БГТУ им. В.Г. Шухова)**



ИНСТИТУТ ИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ И УПРАВЛЯЮЩИХ СИСТЕМ

Лабораторная работа №7

по дисциплине: Базы данных

тема: «Организация взаимодействия с базой данных через приложение, использующее технологию ORM»

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Лабораторная работа №7

Цель работы: разработать приложение, использующее технологию ORM, для взаимодействия с базой данных.

Вариант 8

Задание к работе

1. Изучить технологию ORM.
2. Разработать снабженное графическим интерфейсом приложение, которое обеспечит подключение к базе данных, разработанной на основе предыдущих лабораторных работ, а также обеспечит выполнение запросов с использованием технологии ORM.

models.py

```
from django.db import models

class House(models.Model):
    address = models.CharField(max_length=256, null=False, blank=False)

class Work(models.Model):
    house = models.ForeignKey(
        House,
        on_delete=models.CASCADE,
        related_name='works',
        null=False
    )
    name = models.CharField(max_length=256, null=False, blank=False)

class Contractor(models.Model):
    name = models.CharField(max_length=256, null=False, blank=False)
    specialization = models.CharField(max_length=256, null=False, blank=False)

class WorkContractor(models.Model):
    work = models.ForeignKey(
        Work,
        on_delete=models.CASCADE,
        related_name='work_contractors',
        null=False
    )
    contractor = models.ForeignKey(
        Contractor,
        on_delete=models.CASCADE,
        related_name='contractor_works',
        null=False
    )

class Resident(models.Model):
    name = models.CharField(max_length=256, null=False, blank=False)
    room = models.BigIntegerField(null=False, blank=False)
    house = models.ForeignKey(
```

```

House,
on_delete=models.CASCADE,
related_name='residents',
null=False
)

```

```

class Bill(models.Model):
    amount = models.BigIntegerField(null=False, blank=False)
    bill_date = models.DateField(null=False, blank=False)
    resident = models.ForeignKey(
        Resident,
        on_delete=models.CASCADE,
        related_name='bills',
        null=False
    )

```

```

class Payment(models.Model):
    amount = models.BigIntegerField(null=False, blank=False)
    payment_date = models.DateField(null=False, blank=False)
    bill = models.ForeignKey(
        Bill,
        on_delete=models.CASCADE,
        related_name='payments',
        null=False
    )

```

serializers.py

```

from .models import *
from rest_framework import serializers

```

```

class HouseSerializer(serializers.ModelSerializer):
    class Meta:
        model = House
        fields = ['id', 'address']

```

```

class WorkSerializer(serializers.ModelSerializer):
    house = HouseSerializer(read_only=True)
    house_id = serializers.PrimaryKeyRelatedField(
        queryset=House.objects.all(),
        source='house'
    )

```

```

class Meta:
    model = Work
    fields = ['id', 'name', 'house_id', 'house']

```

```

class ContractorSerializer(serializers.ModelSerializer):
    class Meta:
        model = Contractor
        fields = ['id', 'name', 'specialization']

```

```

class WorkContractorSerializer(serializers.ModelSerializer):
    work = WorkSerializer(read_only=True)
    contractor = ContractorSerializer(read_only=True)
    work_id = serializers.PrimaryKeyRelatedField(
        queryset=Work.objects.all(),
        source='work'
    )
    contractor_id = serializers.PrimaryKeyRelatedField(

```

```

        queryset=Contractor.objects.all(),
        source='contractor'
    )

    class Meta:
        model = WorkContractor
        fields = ['id', 'work_id', 'work', 'contractor_id', 'contractor']

```

```

class ResidentSerializer(serializers.ModelSerializer):
    house = HouseSerializer(read_only=True)
    house_id = serializers.PrimaryKeyRelatedField(
        queryset=House.objects.all(),
        source='house'
    )

```

```

    class Meta:
        model = Resident
        fields = ['id', 'name', 'room', 'house_id', 'house']

```

```

class BillSerializer(serializers.ModelSerializer):
    resident = ResidentSerializer(read_only=True)
    resident_id = serializers.PrimaryKeyRelatedField(
        queryset=Resident.objects.all(),
        source='resident'
    )

```

```

    class Meta:
        model = Bill
        fields = ['id', 'amount', 'bill_date', 'resident_id', 'resident']

```

```

class PaymentSerializer(serializers.ModelSerializer):
    bill = BillSerializer(read_only=True)
    bill_id = serializers.PrimaryKeyRelatedField(
        queryset=Bill.objects.all(),
        source='bill'
    )

```

```

    class Meta:
        model = Payment
        fields = ['id', 'amount', 'payment_date', 'bill_id', 'bill']

```

views.py

```

from django.shortcuts import render
from rest_framework import viewsets, status
from rest_framework import permissions
from rest_framework.response import Response

```

```

from .serializers import *
from .models import *

```

```

class HouseViewSet(viewsets.ModelViewSet):
    queryset = House.objects.all()
    serializer_class = HouseSerializer

```

```

class WorkViewSet(viewsets.ModelViewSet):
    queryset = Work.objects.all()
    serializer_class = WorkSerializer

```

```

class ContractorViewSet(viewsets.ModelViewSet):
    queryset = Contractor.objects.all()
    serializer_class = ContractorSerializer

```

```

class WorkContractorViewSet(viewsets.ModelViewSet):
    queryset = WorkContractor.objects.all()

```

```
serializer_class = WorkContractorSerializer
```

```
class ResidentViewSet(viewsets.ModelViewSet):  
    queryset = Resident.objects.all()  
    serializer_class = ResidentSerializer
```

```
class BillViewSet(viewsets.ModelViewSet):  
    queryset = Bill.objects.all()  
    serializer_class = BillSerializer
```

```
class PaymentViewSet(viewsets.ModelViewSet):  
    queryset = Payment.objects.all()  
    serializer_class = PaymentSerializer
```

urls.py

```
from django.contrib import admin  
from django.urls import path, include  
from rest_framework import routers  
from my_app import views
```

```
router = routers.DefaultRouter()  
router.register(r'house', views.HouseViewSet)  
router.register(r'work', views.WorkViewSet)  
router.register(r'contractor', views.ContractorViewSet)  
router.register(r'work_contractor', views.WorkContractorViewSet)  
router.register(r'resident', views.ResidentViewSet)  
router.register(r'bill', views.BillViewSet)  
router.register(r'payment', views.PaymentViewSet)
```

```
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path("", include(router.urls)),  
    path('api-auth/', include('rest_framework.urls', namespace='rest_framework')),  
]
```

Для запуска проекта необходимо выполнить:

```
python3 manage.py makemigrations my_app
```

```
python3 manage.py migrate
```

```
python3 manage.py runserver
```

Вывод данных:

GET

http://127.0.0.1:8000/house/

Params

Auth

Headers (9)

Body

Scripts

Body

{}

JSON

Preview

Visualize

```
1  [
2    {
3      "id": 1,
4      "address": "Tokyo 1"
5    },
6    {
7      "id": 2,
8      "address": "Moscow"
9    },
10   {
11     "id": 3,
12     "address": "London"
13   },
14   {
15     "id": 4,
16     "address": "New York"
17   }
18 ]
```

GET

http://127.0.0.1:8000/work/

Params

Auth

Headers (9)

Body

Scripts

Body

{}

JSON

Preview

Visualize

```
1  [
2    {
3      "id": 1,
4      "name": "Build 1",
5      "house_id": 1,
6      "house": {
7        "id": 1,
8        "address": "Tokyo 1"
9      }
10   },
11   {
12     "id": 2,
13     "name": "Draw 1",
14     "house_id": 1,
15     "house": {
16       "id": 1,
17       "address": "Tokyo 1"
18     }
19   }
20 ]
```

GET

http://127.0.0.1:8000/contractor/

Params

Auth

Headers (9)

Body

Scripts

Body

{}

JSON

Preview

Visualize

```
1  [
2    {
3      "id": 1,
4      "name": "Petr",
5      "specialization": "Builder"
6    },
7    {
8      "id": 2,
9      "name": "Petr 2",
10     "specialization": "Builder 2"
11   },
12   {
13     "id": 3,
14     "name": "Vlad",
15     "specialization": "Drawer"
16   },
17   {
18     "id": 4,
19     "name": "Vlad 2"
20   }
21 ]
```

GET

http://127.0.0.1:8000/work_contractor/

Params

Auth

Headers (9)

Body

Scripts

Settings

Body

{}

JSON

Preview

Visualize

```
1  [
2    {
3      "id": 1,
4      "work_id": 1,
5      "work": {
6        "id": 1,
7        "name": "Build 1",
8        "house_id": 1,
9        "house": {
10         "id": 1,
11         "address": "Tokyo 1"
12       }
13     },
14     "contractor_id": 1,
15     "contractor": {
16       "id": 1,
17       "name": "Petr",
18       "specialization": "Builder"
19     }
20   },
21   {
22     "id": 2,
23     "work_id": 2,
24     "work": {
25       "id": 2,
26       "name": "Draw 1",
27       "house_id": 1,
28       "house": {
29         "id": 1,
30         "address": "Tokyo 1"
31       }
32     },
33     "contractor_id": 2,
34     "contractor": {
35       "id": 2,
36       "name": "Petr 2",
37       "specialization": "Builder 2"
38     }
39   },
40   {
41     "id": 3,
42     "work_id": 3,
43     "work": {
44       "id": 3,
45       "name": "Drawer",
46       "house_id": 1,
47       "house": {
48         "id": 1,
49         "address": "Tokyo 1"
50       }
51     },
52     "contractor_id": 3,
53     "contractor": {
54       "id": 3,
55       "name": "Vlad",
56       "specialization": "Drawer"
57     }
58   },
59   {
60     "id": 4,
61     "work_id": 4,
62     "work": {
63       "id": 4,
64       "name": "Drawer 2",
65       "house_id": 1,
66       "house": {
67         "id": 1,
68         "address": "Tokyo 1"
69       }
70     },
71     "contractor_id": 4,
72     "contractor": {
73       "id": 4,
74       "name": "Vlad 2",
75       "specialization": "Drawer 2"
76     }
77   }
78 ]
```

GET http://127.0.0.1:8000/resident/

Params Auth Headers (9) Body ● Scripts ● Settings

Body ▾

{ } JSON ▾ Preview Visualize ▾

```
1  [
2    {
3      "id": 1,
4      "name": "Vlad 1",
5      "room": 1,
6      "house_id": 1,
7      "house": {
8        "id": 1,
9        "address": "Tokyo 1"
10     },
11   },
12   {
13     "id": 2,
14     "name": "Oleg",
15     "room": 2,
16     "house_id": 3,
17     "house": {
18       "id": 3,
19       "address": "London"
20     }
21   }
22 ]
```

GET http://127.0.0.1:8000/bill/

Params Auth Headers (9) Body ● Scripts ● Settings

Body ▾

{ } JSON ▾ Preview Visualize ▾

```
1  [
2    {
3      "id": 1,
4      "amount": 3000,
5      "bill_date": null,
6      "resident_id": 4,
7      "resident": {
8        "id": 4,
9        "name": "Petr",
10       "room": 3,
11       "house_id": 2,
12       "house": {
13         "id": 2,
14         "address": "Moscow"
15       }
16     },
17   },
18   {
19     "id": 2,
20     "amount": 1500,
21   }
22 ]
```

GET http://127.0.0.1:8000/payment/

Params Auth Headers (9) Body ● Scripts ● Settings

Body ▾

{ } JSON ▾ Preview Visualize ▾

```
1  [
2    {
3      "id": 1,
4      "amount": 100,
5      "payment_date": null,
6      "bill_id": 4,
7      "bill": {
8        "id": 4,
9        "amount": 100,
10       "bill_date": null,
11       "resident_id": 1,
12       "resident": {
13         "id": 1,
14         "name": "Vlad 1",
15         "room": 1,
16         "house_id": 1,
17         "house": {
18           "id": 1,
19           "address": "Tokyo 1"
20         }
21       }
22     }
23 ]
```